

Howard University
College of Engineering & Architecture
Department of Electrical Engineering & Computer Science
EECE 306-01: Fundamentals of Electromagnetics Laboratory
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Lab 6: Electric Potential, the Gradient, and Path Independence

Objectives

By the end of this session you will have

- implemented the potential, the numerical gradient and the closed path integrator,
- cross-checked two independent routes to the same field,
- verified in three geometries that the electrostatic field is conservative.

1. A second route to the same physics

The element list you built in Lab 3 already contains everything needed to compute the potential. Only the kernel changes, from $\vec{r}/|\vec{r}|^3$ to $1/|\vec{r}|$,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^M \frac{q_i w_i}{|\vec{r} - \vec{r}_i|}$$

Having two independent routes to the same physical content is not redundancy. It is the basis of today's first test, because $\vec{E}$ computed directly and $\vec{E}$ computed as $-\nabla V$ must agree, and both come from your own code.

1.1 A caution about the reference point

The expression above sets $V = 0$ at infinity. That choice is only available when the total charge occupies a finite region. A source built to approximate an infinite line has no finite potential relative to infinity, and the sum above will grow without bound as elements are added.

Your `em.field.V` should detect this situation and issue a warning rather than return a large number silently. A practical test is to compare the potential computed with N elements against the potential with $2N$ elements over the same geometry. For a finite source the two agree. For a source whose extent grows with N , they do not.

State in your help block what reference your potential uses. A potential without a stated reference is not a number, it is half of a number.

2. Part I. Consistency, $\vec{E} = -\nabla V$

Implement `em.op.grad(f, r, h)` by central differences on each Cartesian component.

Now compute the field of a multi-charge source two ways over a plane, once with `em.field.E` and once as the negative gradient of `em.field.V`. Produce an error map, the magnitude of the difference as a filled contour over the plane.

What a disagreement tells you

Both numbers come from your own code, so this test cannot tell you that your physics is right. What it can do is localize a bug, which is often more useful.

An error map that is uniformly small everywhere except near the sources tells you the finite difference step is too large relative to how fast the field varies there, which is a discretization issue and not a bug. An error map that is large everywhere tells you one of the two kernels has a constant wrong, most likely a factor of 4π or a sign. An error map with structure aligned to one coordinate axis tells you the gradient is differencing the wrong component.

Reading the shape of an error is a skill, and it is faster than reading the code.

3. Part II. Choosing the step size

A central difference has two competing error sources. The truncation error falls as h^2 , so a smaller h looks better. The round-off error grows as $\varepsilon_{\text{mach}}/h$, because the numerator subtracts two nearly equal numbers, so a larger h looks better. The total error has a minimum, and for double precision the best h is around 10^{-5} to 10^{-6} , which is why the API sets the default `h` at 10^{-5} m.

The practical consequence is the one to remember. Do not drive h to the smallest number you can, expecting a better derivative. Below roughly 10^{-6} the answer gets worse, not better. The default is a good choice for the fields in this course, and you only need to revisit it if a gradient comes out visibly noisy.

4. Part III. The field is conservative

Implement `em.field.circulation(F, curve, tspan, N, rule)`, the closed path counterpart of last week's flux integrator. Like it, it takes a field function handle.

$$\oint_C \vec{E} \cdot d\vec{l} = 0$$

This must hold for *every* closed path in *every* electrostatic field. Verify it across a grid of cases, three paths crossed with three sources.

Paths	Sources
A circle	Several point charges
A square	A finite line charge
A closed path of your own choosing	A volume charge

Use a path of your own that is deliberately awkward, not centered on anything and not lying in a coordinate plane. A test that only passes for tidy configurations has not tested much.

This used to be the final project

In the previous version of this course, demonstrating numerically that the electrostatic field is conservative was one of the offered final project topics, worth several weeks of work. Today it is one exercise in a two hour lab.

Nothing about the physics got easier. What changed is that the field engine already existed from Lab 3, the quadrature already existed from Lab 4, and the path integrator is forty lines because it reuses both. That is the entire argument for building cumulatively, and it is worth

pausing on, because the same compounding is about to make Labs 9 and 10 much cheaper than they look.

5. Part IV. Seeing the potential

Implement `em.viz.contourV(f, xlim, ylim, N, opts)`, a filled contour of a scalar field on a plane.

Plot the equipotentials of a dipole, then overlay the field lines from `em.viz.quiver2` with normalization on. Field lines and equipotentials must cross at right angles everywhere, which is a visual restatement of $\vec{E} = -\nabla V$, since a gradient is always perpendicular to a level set.

Include the figure and say in one sentence what would be visibly wrong if your gradient had a sign error.

Required tests

Write `tests/test_lab06.m` containing at least these checks.

1. Potential of a point charge matches $Q/(4\pi\epsilon_0 r)$ at ten radii.
2. $\vec{E}$ from `em.field.E` and $\vec{E}$ from $-\nabla V$ agree at points not too close to a source.
3. Circulation of $\vec{E}$ vanishes on all nine path and source combinations, with a threshold small compared with the field magnitude.
4. Circulation of a deliberately non-conservative test field, for example $\vec{F} = -y\hat{x} + x\hat{y}$, is *not* zero on a circle. This confirms the integrator can detect a failure and is not simply returning zero.
5. Potential difference between two points computed from `em.field.V` agrees with the negative line integral of $\vec{E}$ along a path joining them.
6. `em.field.V` warns when handed a source whose extent grows with the element count.

Item 5 deserves attention. A test that only ever confirms zero cannot distinguish correct code from code that always returns zero. Every conservation test you write should be paired with a case that ought to fail.

Deliverables

Submit through Canvas by the start of the next lab session.

1. **Toolbox snapshot**, `TeamNN_Lab06.zip`.
2. **Lab notebook**, published script exported to PDF, three to five pages, with the error map and the equipotential figure with field lines overlaid.
3. **Test output** pasted into the notebook.

Looking ahead

Next week you stop summing over sources and start solving a differential equation. Read Sections 2.9 and 2.10 of Wentworth before you come, and be ready to derive a finite difference stencil on the board.